HUANG XINJIE

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EDUCATION

University of Science and Technology of China Hefei, Anhui, China
School of the Gifted Young
B.S in Physics September 2023 – present

- GPA: 3.87/4.30 (23/108)
- Selected awards: Excellent Student Scholarship – Gold (the top 3% of their class in the annual comprehensive evaluation)
- Core course: Function of Complex Variable A; Probability Theory and Mathematical Statistics; Electrodynamics; Theoretical Mechanics A; Optics A; Quantum Mechanics A; Computational Method

HONORS AND REWARD

- Excellent Student Scholarship – Gold (2025): recognizes students in the top 3% of their class in the annual comprehensive evaluation
- Rose Fund Endeavor Scholarship (2025): recognizes students who have made outstanding improvements in their academic progress year-over-year
- First Prize in Mathematics competition of Chinese College Students (2024)
- First Prize in Chinese Physics Olympiad (2022)

RESEARCH EXPERIENCE

University of Science and Technology of China (Department of Physics) Hefei, Anhui, China
Doing program under the supervision of Professor Tao Xiaoping October 2025 –Present
Automatically adapts to wireless charging

- Using cameras to locate the electric appliances and adapt the direction of charging coil automatically
- Learning how to deal with the Single-Chip Microcomputer and other devices such as Stepper motor
- Gaining experience of image recognition and control of electromechanical components

Hong Kong University of Science and Technology (Department of Physics) Hong Kong, China
Research Assistant to Prof. Hyokeun Park June 2025 – August 2025
Single-molecule dynamics of motor proteins in living cells

- Performed TIRF microscopy to visualize single myosin VI motor proteins in live cells
- Conducted cell culture, fluorescent transfection, and sample preparation
- Developed an automated dual-channel image registration algorithm, achieving superior alignment accuracy compared to manual approaches
- Gained extensive hands-on experience in biophysics research, transitioning from theoretical physics to experimental biological imaging

University of Science and Technology of China (Department of Unclear Science) Hefei, Anhui, China
Research Assistant to Assoc. Prof. Z.X. Liu October 2024 –November 2025
The simulation analysis and prediction of the Impact of Energy Particles on the Pedestal instabilities of tokamak

- Performed tokamak plasma simulations using BOUT++ and high-performance Linux computing environments
- Built automated scripts to manage large-scale simulations and analyze physical parameters
- Investigated the influence of additional current on pedestal stability and performed predictive modeling for future states of the plasma
- Strengthened programming and computational modeling skills in an interdisciplinary physics environment

ADDITIONAL INFORMATION

Rural Volunteer Service to Liuzhi, Guizhou

- Learning the development of Liuzhi
- Teaching science in the local primary school

Computer Skills

- Proficient in python, C, MATLAB
- Experience in Automation and Data analysis, on everywhere of my life